

Git Commands Guide for Beginners

1. Check Git Installation

Command:

```
git --version
```

Example Output:

```
git version 2.48.1
```

Purpose:

Verifies that Git is installed on your machine.

2. Configure Git

Commands:

```
git config --global user.name "John Doe"
```

```
git config --global user.email "john@example.com"
```

Example:

After configuration, every commit you make will be associated with these details.

3. Create a New Repository

Commands:

```
mkdir MyProject
```

```
cd MyProject
```

```
git init
```

Example:

You create a folder called MyProject and initialize Git tracking for it.

4. Check Repository Status

Command:

```
git status
```

Example:

If app.txt is newly created, Git will display it under "Untracked files".

5. Add Files to Staging Area

Commands:

```
git add app.txt
```

```
git add .
```

Example:

app.txt is prepared for the next commit.

6. Commit Changes

Command:

```
git commit -m "Added app.txt file"
```

Example:

A snapshot of the current project state is saved in Git history.

7. View Commit History

Commands:

```
git log
```

```
git log --oneline
```

Example Output:

```
a12bc34 Added app.txt file
```

```
b45de67 Initial commit
```

8. View Changes

Command:

```
git diff
```

Example:

Shows the exact lines added, modified, or removed before committing.

9. Create and Switch Branches

Command:

```
git checkout -b feature-login
```

Example:

You start developing a login feature without affecting the main branch.

10. Merge a Branch

Commands:

```
git checkout main
```

```
git merge feature-login
```

Example:

All completed login feature changes are merged into the main branch.

11. Connect to GitHub

Command:

```
git remote add origin https://github.com/user/myproject.git
```

Example:

Links your local repository with a GitHub repository.

12. Push Code to GitHub

Command:

```
git push -u origin main
```

Example:

Uploads local commits to GitHub for backup and collaboration.

13. Clone an Existing Repository

Command:

```
git clone https://github.com/user/project.git
```

Example:

Downloads a project from GitHub to your local machine.

14. Pull Latest Changes

Command:

```
git pull
```

Example:

Downloads and merges the latest code changes from GitHub.

15. Undo Staged Changes

Command:

```
git restore --staged app.txt
```

Example:

Removes app.txt from staging while keeping file changes.

16. Discard Local Changes

Command:

```
git restore app.txt
```

Example:

Reverts app.txt to its last committed state.

17. Real Project Workflow Example

Step 1: git pull

Step 2: git checkout -b feature-login

Step 3: Make code changes

Step 4: git add .

Step 5: git commit -m "Added login functionality"

Step 6: git push origin feature-login

This is a common workflow used by development teams.

18. Git Cheat Sheet

git init

git status

git add .

git commit -m "message"

git log --oneline

git diff

git checkout -b branch-name

git merge branch-name

git push

git pull

git clone URL